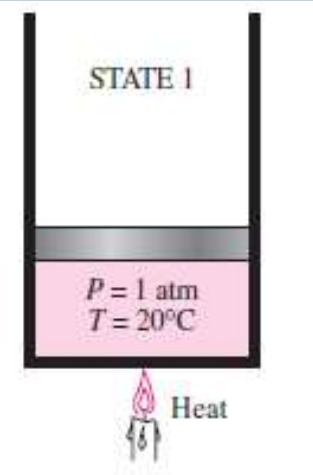
Unit - VI

PROPERTIES OF STEAM

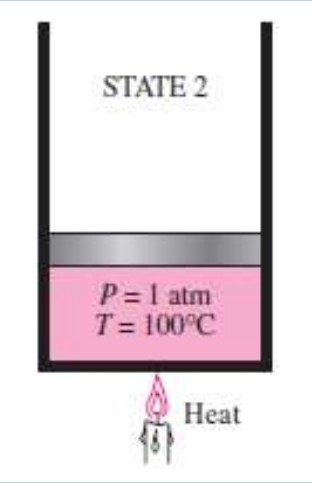
Compressed Liquid and Saturated Liquid

Liquid that is not about to vaporize, is called Compressed liquid.

Liquid that is about to vaporize, is called Saturated liquid.



At 1 atm and 20°C, water exists in the liquid phase (*compressed liquid*).



At 1 atm pressure and 100°C, water exists as a liquid that is ready to vaporize (*saturated liquid*).

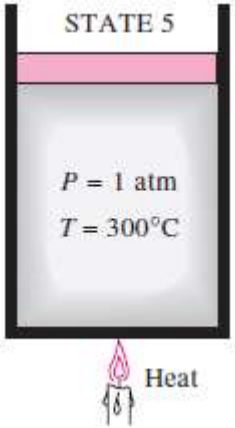
Saturated Vapor and Superheated Vapor

A vapour that is about to condense is called Saturated vapor.

A vapor that is not about to condense (i.e., not a saturated vapor) is called a superheated vapor.



At 1 atm pressure, the temperature remains constant at 100°C until the last drop of liquid is vaporized (*saturated vapor*).

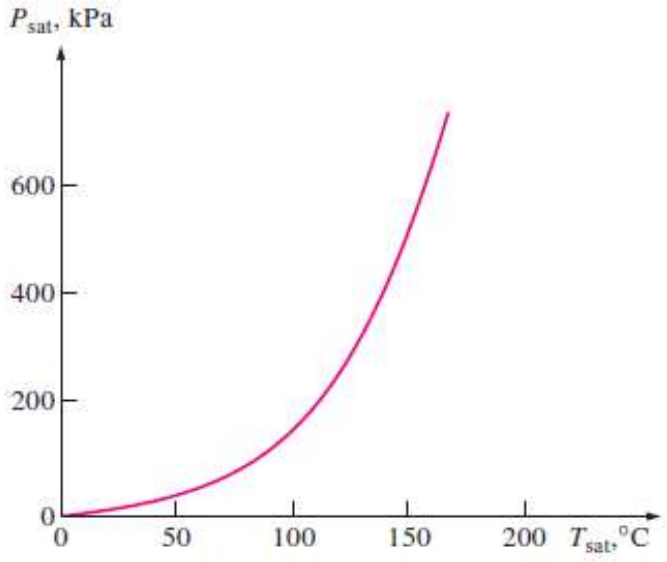


As more heat is transferred, the temperature of the vapor starts to rise (*superheated vapor*).

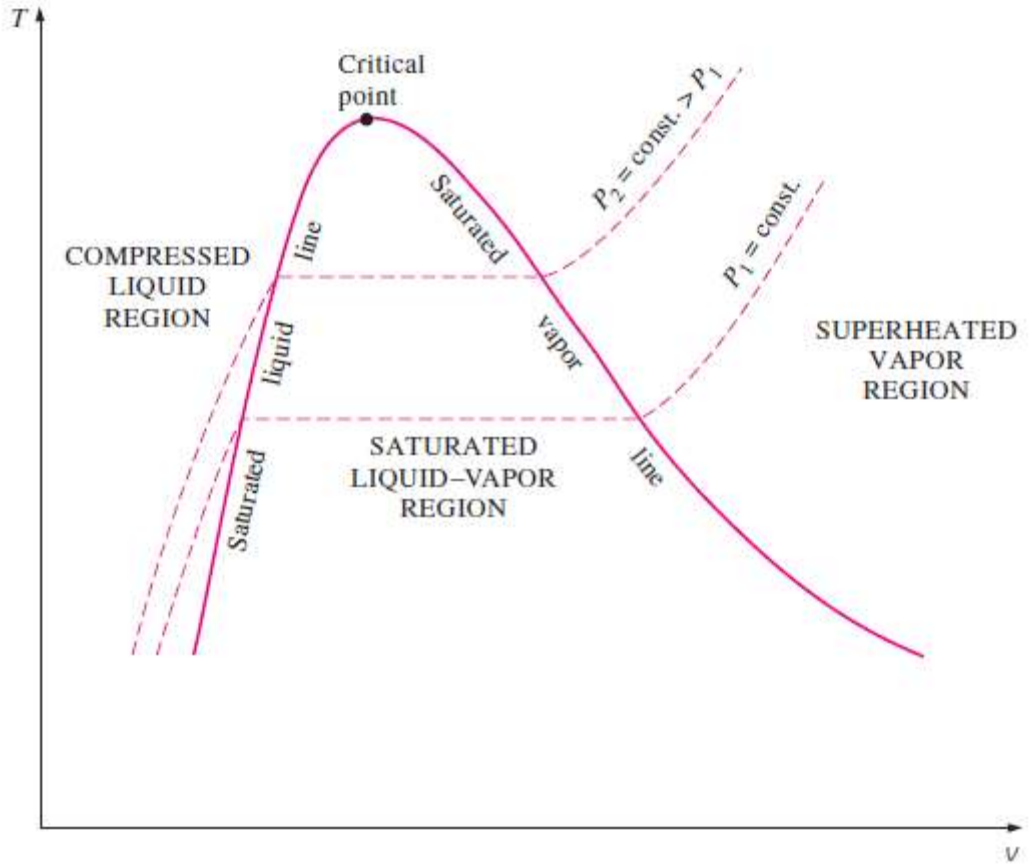
Saturation Temperature and Saturation Pressure

At a given pressure, the temperature at which a pure substance changes phase is called the saturation temperature ***Tsat.***

Likewise, at a given temperature, the pressure at which a pure substance changes phase is called the saturation pressure ***Psat.***

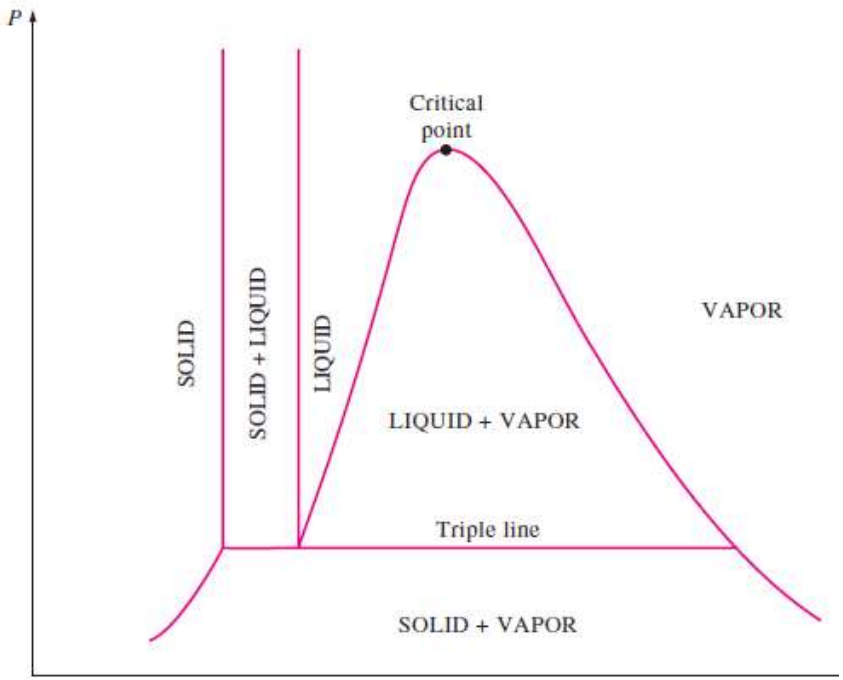


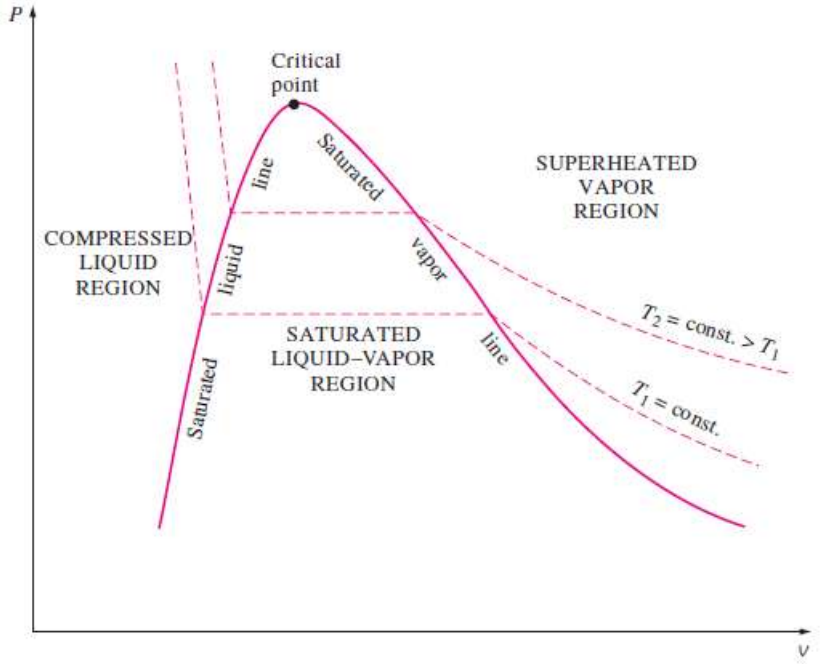
PROPERTY DIAGRAMS FOR PHASE-CHANGE PROCESSES

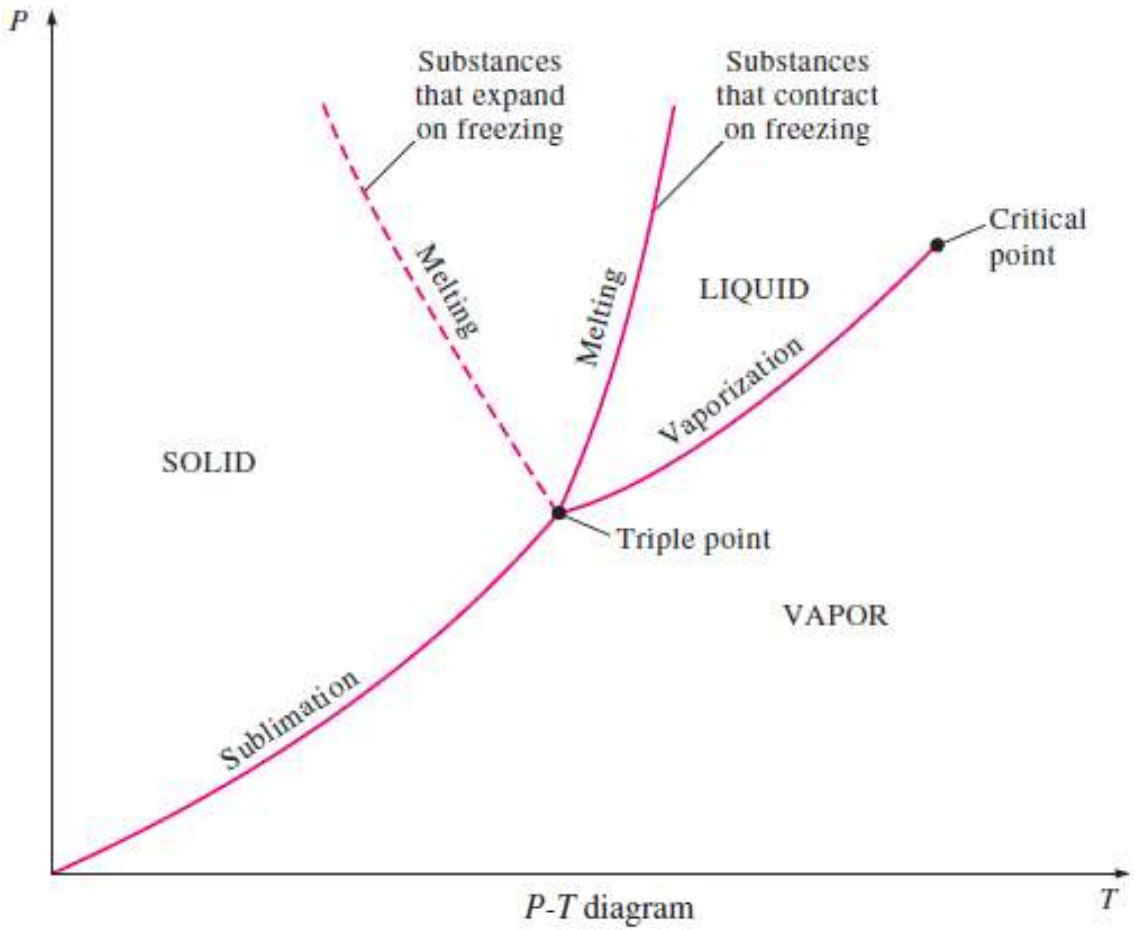


T-*v* diagram

P-V diagram

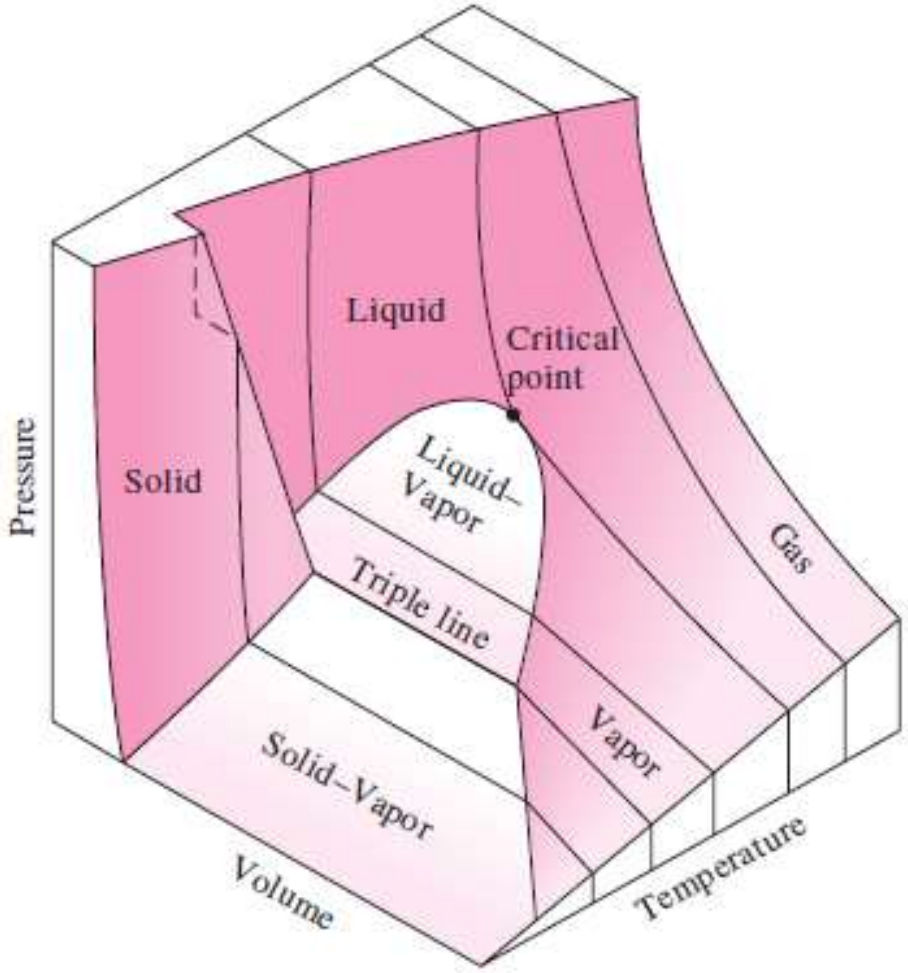

P
Critical point
SOLID
SOLID + LIQUID
LIQUID
VAPOR
LIQUID + VAPOR
Triple line
SOLID + VAPOR


P
Critical point
Saturated liquid line
Saturated vapor line
SUPERHEATED VAPOR REGION
COMPRESSED LIQUID REGION
SATURATED LIQUID–VAPOR REGION
T_2 = const. > T_1
T_1 = const.
v


P
Substances
that expand
on freezing
Substances
that contract
on freezing
Critical
point
Melting
Melting
LIQUID
Vaporization
SOLID
Triple point
VAPOR
Sublimation
T

P-T diagram

P-v-T surface


Liquid
Critical point
Solid
Liquid–Vapor
Gas
Triple line
Vapor
Solid–Vapor
Pressure
Volume
Temperature

Boiling Point

The boiling point is the temperature at which the saturated vapour pressure of a liquid becomes equal to the external pressure, causing the liquid to change into vapour throughout its mass.

Example: The boiling point of water at 1 atm pressure is 100°C.

Saturation Point

The saturation point is the state at which a substance is about to undergo a phase change from liquid to vapour or from vapour to liquid at a given pressure and temperature.

Example: Water at 100°C and 1 atm is at its saturation condition.

Saturation Pressure

The saturation pressure is the pressure at which a pure substance changes its phase between liquid and vapour at a given temperature.

Example: The saturation pressure of water at 100°C is approximately 101.325 kPa.

Saturation Temperature

The saturation temperature is the temperature at which a pure substance changes its phase between liquid and vapour at a given pressure.

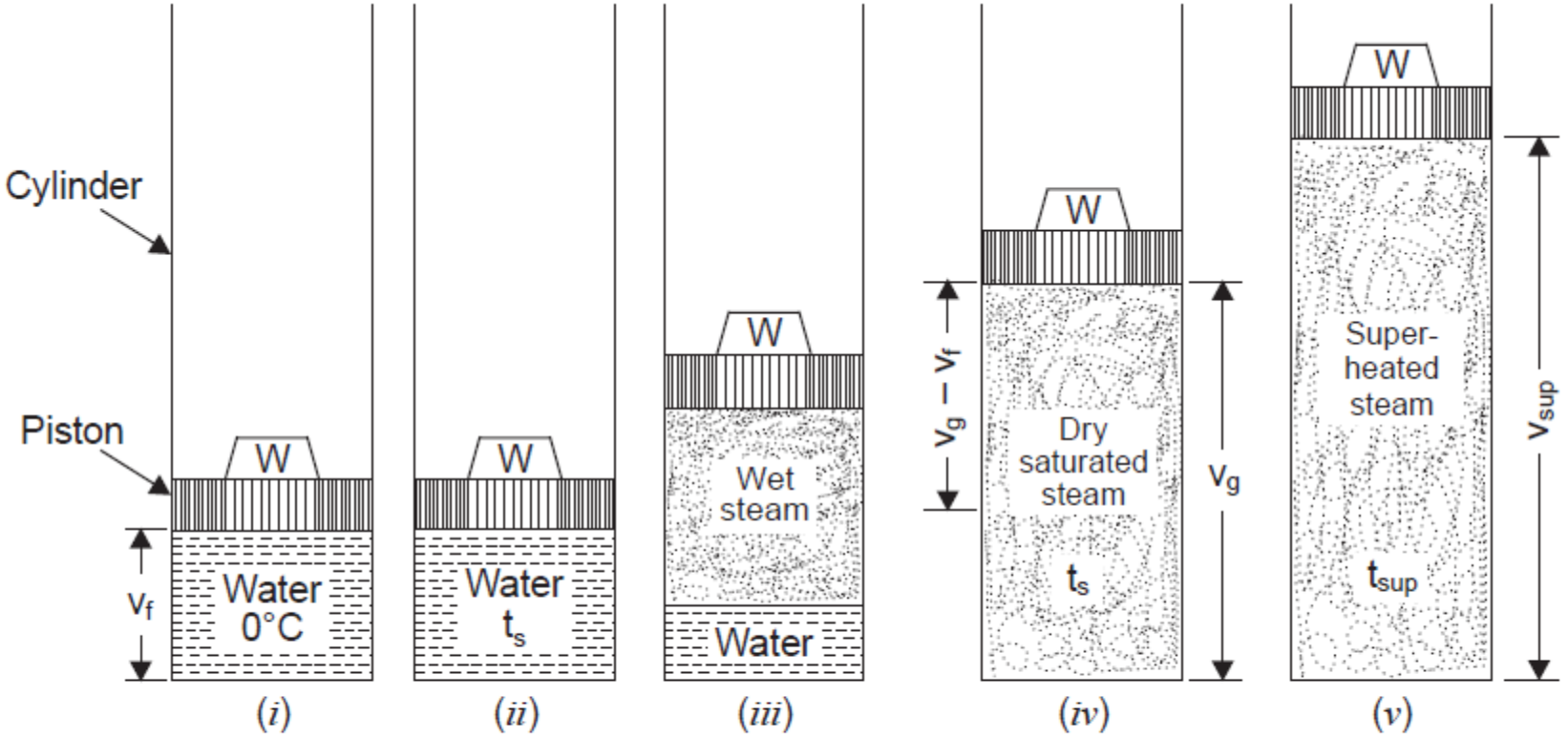
Example: The saturation temperature of water at 101.325 kPa is 100°C.

Triple Point

The triple point is the unique temperature and pressure at which the solid, liquid and vapour phases of a pure substance coexist in thermodynamic equilibrium.

Example: The triple point of water occurs at 0.01°C (273.16 K) and approximately 611.657 Pa.

Formation of steam



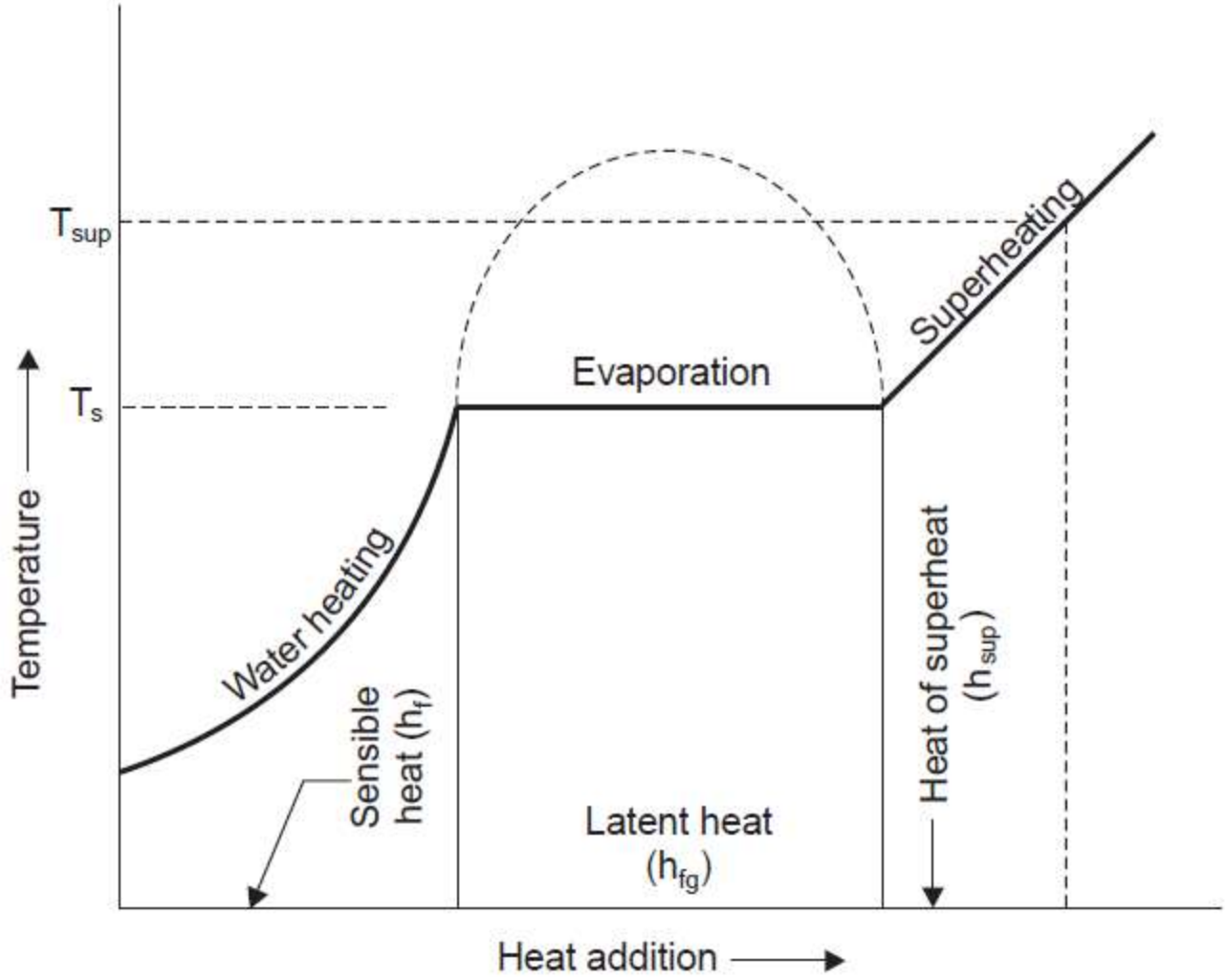
t_s = Saturation temp.

t_{sup} = Temperature of superheated steam

v_f = Volume of water

v_g = Volume of dry and saturated steam

v_{sup} = Volume of superheated steam



Graphical representation of formation of steam

Steam

Steam is a vapour of water, and is invisible when pure and dry. It is used as a working substance in the operation of steam engines and steam turbines. It obeys the laws of gases, when it is perfectly dry.

Wet steam: When the steam contains moisture or particles of water, it is termed as wet steam.

Dry saturated steam: When the steam does not contain any moisture or suspended particles of water, it is termed as dry saturated steam. It is obtained when the wet steam is further heated.

Superheated steam: When the dry saturated steam is further heated at constant pressure thus raising its temperature, it is termed as superheated steam. Because heating of steam at constant pressure therefore volume of superheated steam increases. It may be noted that the volume of 1 kg of superheated steam is greater than the volume of 1 kg of dry saturated steam.

Dryness fraction: It is the ratio of the actual mass of dry steam to the mass of same quantity of wet steam. It is represented by x. Mathematically it is written as,

$$x = \frac{m_g}{m_g + m_f} = \frac{m_g}{m_w}$$

Where,

m_g = *actual mass of dry steam*

m_f = *mass of water*

m_w = *mass of wet steam* = $(m_g + m_f)$

Sensible heat of water: It is defined as the amount of heat absorbed by one kilogram of water, when heated at constant pressure, from 0^oC (freezing point) to saturation temperature (formation of steam). It is also known as liquid heat.

Latent heat of vaporization: It is defined as the amount of heat required to evaporated one kilogram of water at its saturation temperature (boiling point) without change of temperature. It is represented by h_{fg}. Its unit is kJ/kg. The latent heat of vaporization of water or latent heat of steam is 2257 kJ/kg. If the steam is wet with a dryness fraction x, then the heat absorbed by wet steam is xh_{fg}.

$$h_{fg} = (h_g - h_f)$$

Enthalpy: It is defined as the amount of heat absorbed by water from 0^oC (freezing point) to saturation point (sensible heat) plus heat absorbed during evaporation (latent heat). It is represented by h_g.

So that,

$$Enthalpy = sensible\ heat + latent\ heat$$

Some expressions for enthalpy,

Wet steam: Enthalpy of wet steam is given by;

$$h = h_f + xh_{fg}$$

Dry steam: Enthalpy of dry steam is given by;

$$h = h_g = h_f + h_{fg} \quad \ldots\ldots\ldots\ldots since\ x = 1\ for\ dry\ air.$$

Superheated steam: From the definition superheated steam it may be written as,

$$h_{sup} = total\ heat\ for\ dry\ steam + heat\ for\ superheated\ steam$$

$$h_{sup} = h_f + h_{fg} + c_p(T_{sup} - T)$$

$$h_{sup} = h_g + c_p(T_{sup} - T)$$

Where,

$$c_p = specific\ heat\ at\ constant\ pressure$$

$$T_{sup} = temperature\ of\ superheated\ steam$$

$$T = saturation\ temperature\ at\ constant\ pressure.$$

Specific volume: It is the volume occupied by the steam per unit mass at given temperature and pressure. It is represented by v. It is expressed in m^3/kg. The reciprocal of specific volume represents the density.

Some expressions for volumes occupied by the steam,

Wet steam: Consider one kilogram of wet steam of dryness fraction x. It means that this steam will have x kg of dry steam and (1-x) kg of water. Let v_f be the volume of one kg of water then, volume of one kilogram of wet steam is given by,

$$= xv_g + (1 - x)v_f$$

Since v_f is very small in comparison to v_g, so the term $(1 - x)v_f$ may be neglected.

So that,

Volume of one kilogram of wet steam

$$= xv_g \text{ m}^3$$

Or

Specific volume of wet steam

$$v = xv_g \text{ m}^3/\text{kg}$$

Dry steam: As studied earlier in dry steam, the mass of water is zero and dryness fraction is unity. So that

Specific volume of dry steam

$$v = v_g \, \mathrm{m^3/kg}$$

Superheated steam: As studied earlier when the dry saturated steam is further heated at constant pressure thus raising its temperature, it is termed as superheated steam. Because heating of steam at constant pressure therefore volume of superheated steam increases. According to Charles's law

$$\frac{v_{sup}}{T_{sup}} = \frac{v_g}{T}$$

Or

$$v_{sup} = \frac{v_g . T_{sup}}{T}$$

Where,

$$v_{sup} = specific\ volume\ of\ superheated\ steam$$
$$T_{sup} = temperature\ of\ superheated\ steam$$
$$v_g = specific\ volume\ of\ dry\ saturated\ steam$$
$$T = saturation\ temperature.$$

Internal energy: Internal energy of steam is define as the energy stored in the steam, above 0°C (freezing point) of water. It may be obtained by subtracting the work done during evaporation to the enthalpy of steam. It is represented by U. Mathematically it is written as,

$$Internal\ energy\ of\ steam$$
$$= Enthalpy\ of\ steam - Workdone\ during\ evaporation$$

Some expressions for enthalpy,

Wet steam: Internal energy of wet steam is given by;

$$U = h - 100 * P * x * v_g = h_f + xh_{fg} - 100 * P * x * v_g \ \ kJ/kg$$

Dry steam: Internal energy of dry steam is given by;

$$U = h_g - 100 * P * v_g \ \ kJ/kg \quad \ldots\ldots\ldots\ldots since\ x = 1\ for\ dry\ air.$$

Superheated steam: From the definition superheated steam it may be written as,

$$U = h_{sup} - 100 * P * v_g$$
$$U = h_g + c_p\left(T_{sup} - T\right) - 100 * P * v_{sup} \ \ kJ/kg$$

h-s Diagram or Mollier Diagram

From the first and second laws of thermodynamics

$$Tds = dh - vdp$$

or

$$\left(\frac{\partial h}{\partial s}\right)_p = T$$

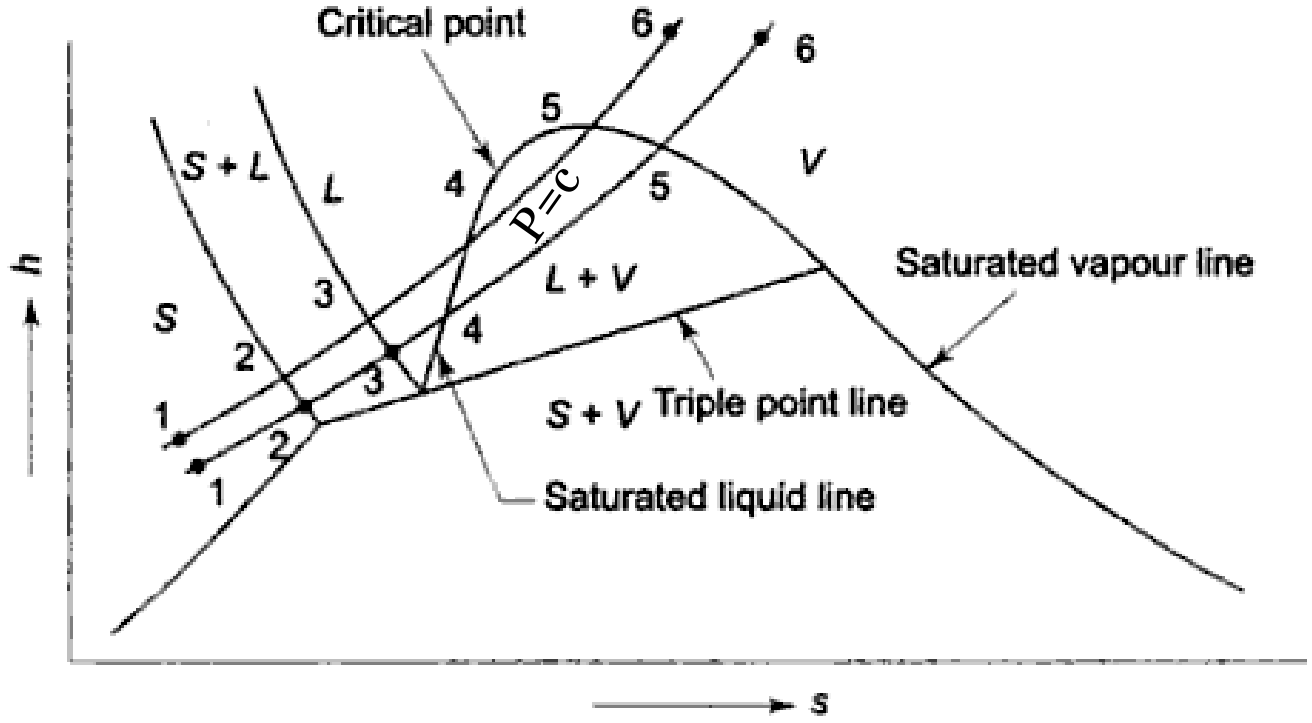


Fig. 9.15 *Phase equilibrium diagram on h-s coordinates (Mollier diagram)*

IMPORTANT TERMS RELATING STEAM FORMATION

- **Sensible heat of water (h_f)** It is defined as the quantity of heat absorbed by 1 kg of water when it is heated from 0°C (freezing point) to boiling point. It is also called total heat (or enthalpy) of water or liquid heat invariably.

- **Latent heat or hidden heat (h_{fg})**. It is the amount of heat required to convert water at a given temperature and pressure into steam at the same temperature and pressure. It is expressed by the symbol h_{fg}.

• **Total heat or enthalpy of wet steam (h).** It is defined as the quantity of heat required to convert 1 kg of water at 0°C into wet steam at constant pressure. It is the sum of total heat of water and the latent heat and this sum is also called enthalpy.

In other words, $h = h_f + xh_{fg}$

If steam is dry and saturated, then $x = 1$ and $h_g = h_f + h_{fg}$.

• **Superheated steam.** When steam is heated after it has become dry and saturated, it is called superheated steam and the process of heating is called superheating. Superheating is always carried out at constant pressure. The additional amount of heat supplied to the steam during superheating is called as '*Heat of superheat*'

The total heat of superheated steam is given by

$$h_{sup} = h_f + h_{fg} + c_{ps}\,(T_{sup} - T_s)$$

Where C_{ps} is the specific heat of superheated steam at constant pressure.

Dryness fraction or Quality (x). The term dryness fraction is related with wet steam. It is defined as the ratio of the mass of water vapour in steam to the mass of steam containing it. It is usually expressed by the symbol ‘x’ or ‘q’.

$$x = \frac{m_g}{m_g + m_l}$$

Where m_g and m_l are the masses of vapour and liquid respectively in the mixture.

The value of x varies between 0 and 1. For saturated water, when water just starts boiling, $x = 0$, and for saturated vapour, when vaporization is complete, $x = 1$, for which the vapour is said to be *dry saturated*.

Let V be the total volume of a liquid vapour mixture of quality x, V_f the volume of the saturated liquid, and V_g the volume of the saturated vapour, the corresponding masses being m, m_f, and m_g respectively.

Now $$m = m_f + m_g$$

and $$V = V_f + V_g$$

$$mv = m_f v_f + m_g v_g$$

$$= (m - m_g)\, v_f + m_g\, v_g$$

$$v = \left(1 - \frac{m_g}{m}\right) v_f + \frac{m_g}{m} v_g$$

$$v = (1 - x)\, v_f + x v_g$$

where $x = \frac{m_g}{m}$, v_f = specific volume of saturated liquid, v_g = specific volume of saturated vapour, and v = specific volume of the mixture of quality x.

Similarly

$$s = (1 - x)\, s_f + x s_g$$
$$h = (1 - x)\, h_f + x h_g$$
$$u = (1 - x)\, u_f + x u_g$$

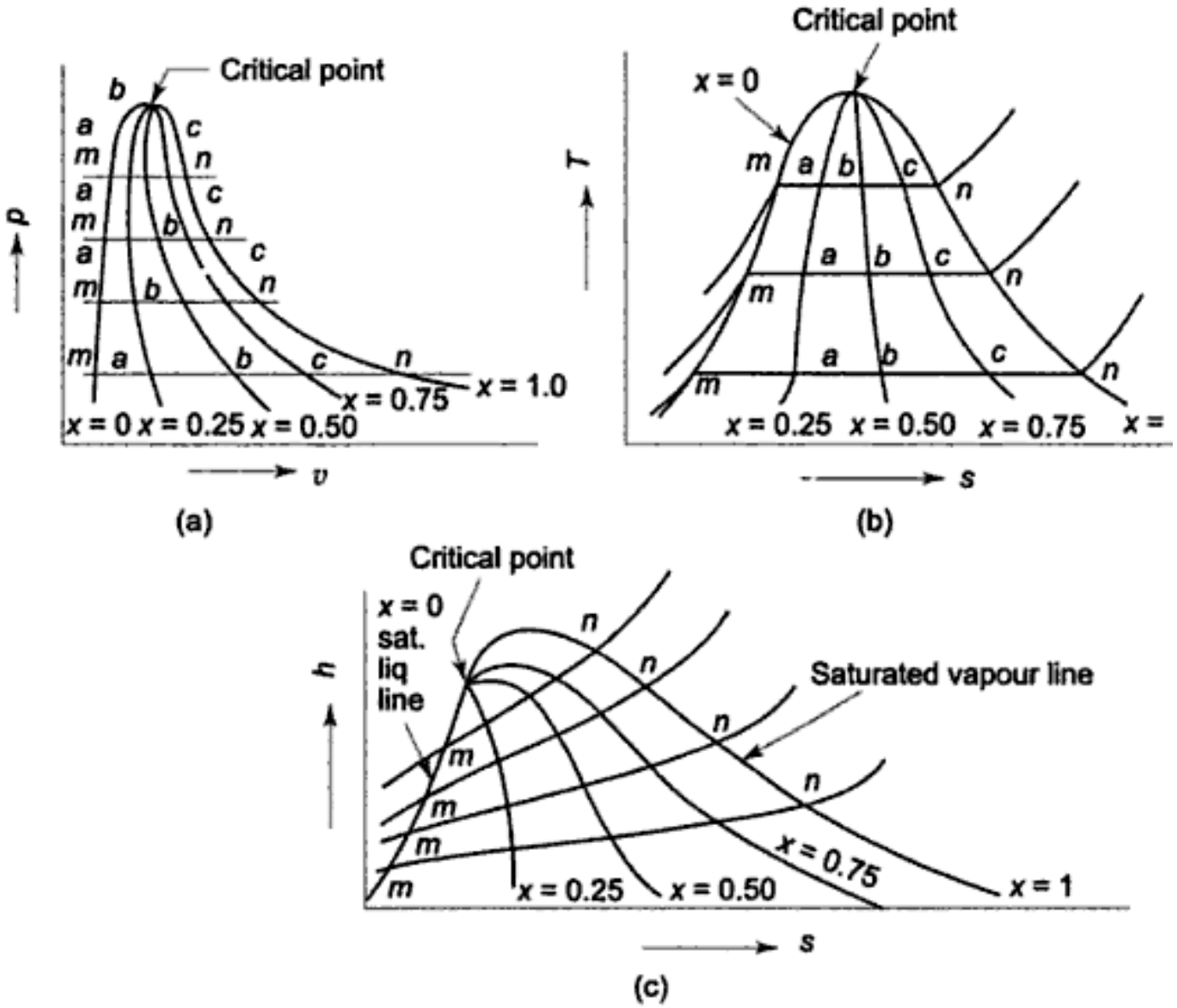
We can also write

$$v = (1 - x)\, v_f + x\, v_g$$
$$= v_f + x(v_g - v_f)$$
$$= v_f + x, v_{fg}$$

or $$v = v_f + x\, v_{fg}$$

Similarly

$$h = h_f + x\, h_{fg}$$
$$s = s_f + x\, s_{fg}$$
$$u = u_f + x\, u_{fg}$$



Constant quality lines

Steam Tables

Steam table is a tabular presentation of properties such as specific enthalpy, entropy, internal energy and specific volume at different saturation pressures and temperatures. Steam table may be on pressure basis or on temperature basis. The table on pressure basis has continuous variation of pressure and corresponding to it : saturation temperature (Tsat), enthalpy of saturated liquid (*hf), enthalpy of saturated vapour (hg),* entropy of saturated liquid (*sf), entropy of saturated vapour (sg), specific volume of saturated liquid (vf),*
specific volume of saturated vapour (*vg), internal energy of saturated liquid (uf), internal energy of* saturated vapour (*ug) are given on unit mass basis.*

Similar to above the temperature based table which gives continuous variation of temperature and corresponding to it saturation pressure and other properties as h_f, h_g, h_{fg}, s_f, s_g, s_{fg}, v_f, v_g, u_f, u_g and u_{fg} are given.

Similarly, steam properties for superheated steam are also estimated and tabulated at some discrete pressures for varying degree of superheat. Super heated steam table are available for getting enthalpy, entropy, specific volume and internal energy separately.

Saturated Water and Steam (Temperature) Tables

Temp. (°C)	*Absolute pressure* (bar)	*Specific enthalpy* (kJ/kg)			*Specific entropy* (kJ/kg K)			*Specific volume* (m^3/kg)	
t	p	h_f	h_{fg}	h_g	s_f	s_{fg}	s_g	v_f	v_g
0	0.0061	−0.02	2501.4	2501.3	−0.0001	9.1566	9.1565	0.0010002	206.3
0.01	0.0061	0.01	2501.3	2501.4	0.000	9.156	9.156	0.0010002	206.2
1	0.0065	4.2	2499.0	2503.2	0.015	9.115	9.130	0.0010002	192.6
2	0.0070	8.4	2496.7	2505.0	0.031	9.073	9.104	0.0010001	179.9
3	0.0076	12.6	2494.3	2506.9	0.046	9.032	9.077	0.0010001	168.1
4	0.0081	16.8	2491.9	2508.7	0.061	8.990	9.051	0.0010001	157.2
5	0.0087	21.0	2489.6	2510.6	0.076	8.950	9.026	0.0010001	147.1
6	0.0093	25.2	2487.2	2512.4	0.091	8.909	9.000	0.0010001	137.7
7	0.0100	29.4	2484.8	2514.2	0.106	8.869	8.975	0.0010002	129.0
8	0.0107	33.6	2482.5	2516.1	0.121	8.829	8.950	0.0010002	120.9
9	0.0115	37.8	2480.1	2517.9	0.136	8.789	8.925	0.0010003	113.4
10	0.0123	42.0	2477.7	2519.7	0.151	8.750	8.901	0.0010004	106.4
11	0.0131	46.2	2475.4	2521.6	0.166	8.711	8.877	0.0010004	99.86
12	0.0140	50.4	2473.0	2523.4	0.181	8.672	8.852	0.0010005	93.78

Saturated Water and Steam (Pressure) Tables

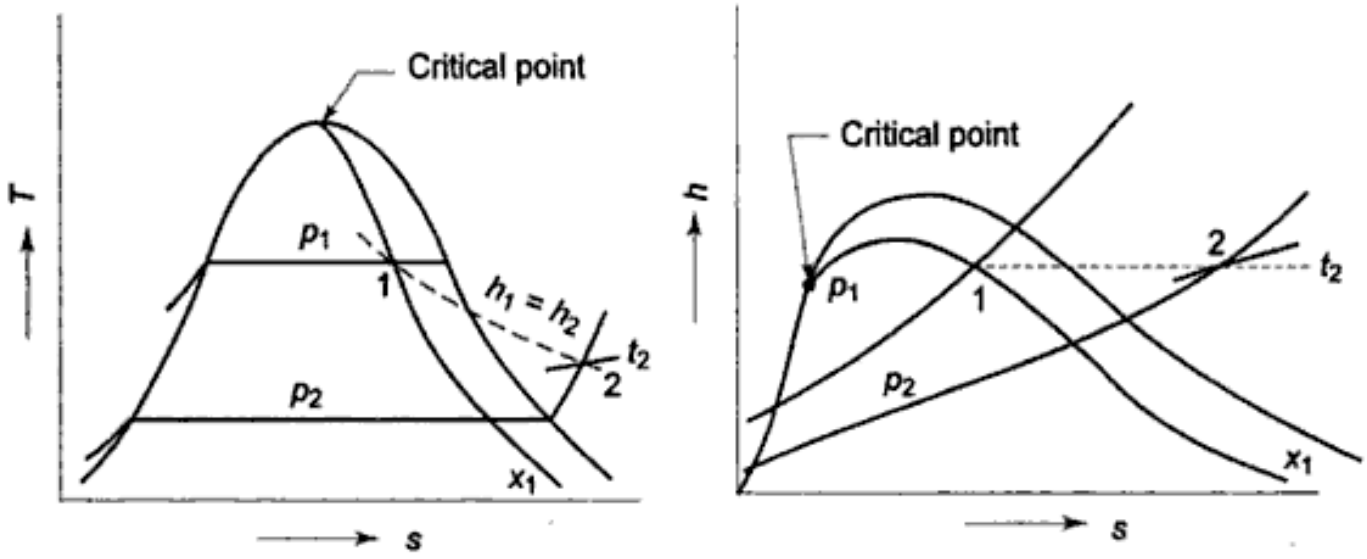
Absolute pressure (bar) p	*Temp. (°C)* t_s	*Specific enthalpy (kJ/kg)* h_f	h_{fg}	h_g	*Specific entropy (kJ/kg K)* s_f	s_{fg}	s_g	*Specific volume (m^3/kg)* v_f	v_g
0.006113	0.01	0.01	2 501.3	2 501.4	0.000	9.156	9.156	0.0010002	206.14
0.010	7.0	29.3	2 484.9	2 514.2	0.106	8.870	8.976	0.0010000	129.21
0.015	13.0	54.7	2 470.6	2 525.3	0.196	8.632	8.828	0.0010007	87.98
0.020	17.0	73.5	2 460.0	2 533.5	0.261	8.463	8.724	0.001001	67.00
0.025	21.1	88.5	2 451.6	2 540.1	0.312	8.331	8.643	0.001002	54.25
0.030	24.1	101.0	2 444.5	2 545.5	0.355	8.223	8.578	0.001003	45.67
0.035	26.7	111.9	2 438.4	2 550.3	0.391	8.132	8.523	0.001003	39.50
0.040	29.0	121.5	2 432.9	2 554.4	0.423	8.052	8.475	0.001004	34.80
0.045	31.0	130.0	2 428.2	2 558.2	0.451	7.982	8.433	0.001005	31.13
0.050	32.9	137.8	2 423.7	2 561.5	0.476	7.919	8.395	0.001005	28.19
0.055	34.6	144.9	2 419.6	2 565.5	0.500	7.861	8.361	0.001006	25.77

Superheated Steam at Various Pressures and Temperatures

↓p (bar) (t_s)	t (°C) →	50	100	150	200	250	300	400	500
	v	149.1	172.2	195.3	218.4	241.5	264.5	310.7	356.8
0.01	u	2445.4	2516.4	2588.4	2661.6	2736.9	2812.2	2969.0	3132.4
(7.0)	h	2594.5	2688.6	2783.6	2880.0	2978.4	3076.8	3279.7	3489.2
	s	9.242	9.513	9.752	9.967	10.163	10.344	10.671	10.960
	v	29.78	34.42	39.04	48.66	48.28	52.9	62.13	71.36
0.05	u	2444.8	2516.2	2588.4	2661.9	2736.6	2812.6	2969.6	3133.0
(32.9)	h	2593.7	2688.1	2783.4	2879.9	2977.6	3076.7	3279.7	3489.2
	s	8.498	8.770	9.009	9.225	9.421	9.602	9.928	10.218

Dryness fraction measurement:

In the measurement of quality, the object is always to bring the state of the substance from the two-phase region to the single-phase or superheated region, where both pressure and temperature are independent. It can be measured either by adiabatic throttling or electric heating.



The initial state (wet) is given by p1 and x1, and the final state by p_2 and t_2 (superheated). Now

since $$h_1 = h_2$$

$$h_{fp1} + x_1 h_{fgp1} = h_2$$

or $$x_1 = \frac{h_2 - h_{fp1}}{h_{fgp1}}$$

With p_2 and t_2 being known, h_2 can be found out from the superheated steam table. The values of h_1 and h_{fgp1} are taken from the saturated steam table corresponding to pressure p_i. Therefore, the quality of the wet steam x_i can be calculated.

$$w_1 h_1 + \boldsymbol{Q = w_1 h_2}$$

where w_1 is the steam flow rate in kg/s ($w_1 = m/t$ kg/s)

$$So,\ h_1 + Q/w_1 = h_2$$

With h_2, Q and w_1 being known, h1 can be computed. Now

$$h_1 = h_{fp1} + x_1 h_{fgp1}$$

Hence x_1 can be evaluated.